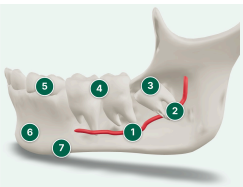
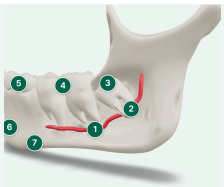
Impacted lower wisdom tooth and the nerve near it

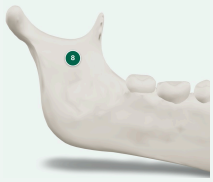
NAGOYA UNIVERSITY



Whole jaw



Roots and nerve



Tongue side

Jaw bone and teeth Nerve, artery and vein (in their canal)

Parts of the model

1 Jaw nerve (inferior alveolar nerve)
- Gives feeling to the lower teeth, gums, lip and chin on this side.
- Runs in a tube in the bone with the inferior alveolar artery and vein. These can also be hurt.
- If the nerve is hurt, the area can feel numb, have pins and needles, feel strange or hurt. This can be short-term or long-term.
- In older adults, damage is more likely, and less likely to heal fully.

2 Roots of the wisdom tooth
- The roots wrap around the jaw nerve.
- So taking the tooth out can hurt the nerve.

3 Wisdom tooth (third molar)
- Impacted: stuck under the gum and bone.
- Unerupted: it has not come through the gum.

4 First and second molars
- Big back teeth used to chew.
- The wisdom tooth sits behind them.

5 First and second premolars
- The teeth in front of the molars.
- They also help to chew.

6 Chin nerve (mental nerve)
- A branch of the jaw nerve. It comes out of the bone through a small hole (the mental foramen).
- Gives feeling to the lower lip and chin.
- If the jaw nerve is hurt further back, the lip and chin can feel different too.

7 Jaw bone (mandible)
- The jaw nerve, artery and vein run in a tube inside this bone.

8 Where the nerve goes into the bone (lingula)
- The jaw nerve, artery and vein go into the bone here, through a hole on the tongue side (the mandibular foramen).
- A small spike of bone at the front edge of this hole is the lingula.

What is the problem?

The wisdom tooth is the last tooth at the back. Its other name is the **third molar**. An **impacted lower wisdom tooth** is stuck under the gum and bone. It has not come through the gum (it is **unerupted**). It cannot grow in the right way.

The nerve

A nerve runs in a tube of bone inside the lower jaw. Here it is called the **jaw nerve**. Its medical name is the **inferior alveolar nerve**. Two blood vessels run with it in the same tube. They are the **inferior alveolar artery and vein**.[1]

The jaw nerve is a **branch** of the **trigeminal nerve**, the main nerve of feeling in the face. It comes from the lowest of its 3 main parts, the **mandibular division**. The nerve, artery and vein go into the jaw bone through a hole on the tongue side. This hole is the **mandibular foramen**.[1] A small spike of bone at its front edge is the **lingula**.[2]

The jaw nerve gives feeling to the lower teeth and gums on the same side.[3] Near the front of the jaw, a branch of it comes out of the bone through a small hole. This branch is the **chin nerve** (**mental nerve**). It gives feeling to the lower lip and chin.[4]

The tongue gets its feeling from a different nerve, the **lingual nerve**. It can also be hurt when a wisdom tooth is taken out.[5] Lasting damage to it is rare. It happens to about 1 to 3 in 1,000 people.[6] The tongue nerve is not shown in this model.

In this model, **the roots of the wisdom tooth wrap around the nerve and artery**.

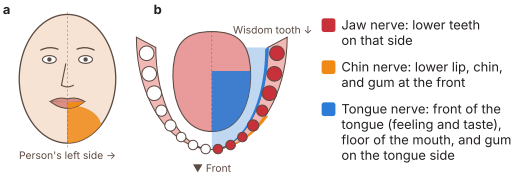
Why taking the tooth out can hurt the nerve

To take the tooth out, the roots must come out of the bone. The roots curve around the nerve. So the nerve can be pressed, stretched or cut as the roots come out. The artery and vein can also be hurt and bleed.[1]

If the jaw nerve is hurt, the chin nerve may also stop working well. This is because it is a branch of the jaw nerve.[4]

What nerve damage feels like

The lower lip, chin, teeth or gums on that side may feel numb, tingle, feel strange or hurt.[3,5]



Where each nerve gives feeling, on the side of the wisdom tooth. If a nerve is hurt, these areas can feel different. **a:** the face from the front. **b:** the lower teeth and tongue, seen from the front.[7,4,8] Drawing made for this study.

Short-term: the feeling comes back, most often within weeks to months. **Long-term:** the feeling does not fully come back.[3]

In older people, the risk is higher and the feeling is less likely to come back fully.[3,9]

How often does it happen?

Some wisdom teeth have roots close to the nerve. A review of 16 studies looked at these teeth. It found:[10]

About 7 in 100
had nerve damage after the whole tooth was taken out

About 1 in 100
had nerve damage that did not go away

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